Practical 3.4: KoboToolbox to CRD Integration

Fetch Tanga flood submissions via API and upload to CRD as spatial layer

Day 3 · 1 hour · Resilience Academy SDI Training

Reference: `crd.resilienceacademy.ac.tz` | Data: Tanga & Mwanza, Tanzania

Learning Objectives:

- Fetch Tanga flood survey data via the KoboToolbox REST API
- Convert GPS coordinates to GeoJSON format using Python
- Upload the resulting layer to your local CRD instance
- Visualize Tanga flood points alongside Tanga satellite imagery in CRD

Overview

This practical integrates two systems: KoboToolbox (data collection) and CRD (geospatial database). We will:

1. Fetch the Tanga flood survey submissions via the KoboToolbox API
2. Convert them to GeoJSON using Python
3. Upload the GeoJSON to the local CRD instance

Step 1: Get Your API Token and Form UID

```
# In KoboToolbox web interface:
# Account Settings > Security > API Key > Show
# Copy this token

# Get your form UID:
# Open your Tanga Flood Survey form
# Look at the URL: http://localhost/api/v2/assets/XXXXXXXXXX/
# The XXXXXXXXXX is your form UID
```

Step 2: Fetch Data via API

```python
#!/usr/bin/env python3
# kobo_to_crd.py --- Tanga flood survey to CRD integration

import requests
import json

# Configuration
KOBO_URL = "http://localhost"
API_TOKEN = "your_token_here" # replace with your token
FORM_UID = "your_form_uid_here" # replace with your form UID

headers = {"Authorization": f"Token {API_TOKEN}"}

# Fetch all submissions
url = f"{KOBO_URL}/api/v2/assets/{FORM_UID}/data/?format=json"
```

```
response = requests.get(url, headers=headers)
data = response.json()
print(f"Total submissions: {data['count']}")

submissions = data['results']
print(f"Fetched {len(submissions)} records")
```

Step 3: Convert to GeoJSON

```
# Convert to GeoJSON FeatureCollection
features = []
for record in submissions:
    # GPS field is typically named like "GPS_Location"
    # Find the GPS field name
    gps_keys = [k for k in record.keys() if 'gps' in k.lower() or 'location' in k.lower()
        ]

    if not gps_keys:
        print(f"No GPS field in record {record['_id']}")
        continue

    gps_key = gps_keys[0]
    gps_value = record.get(gps_key, "")

    if not gps_value:
        continue

    # GPS format: "latitude longitude altitude accuracy"
    parts = str(gps_value).split()
    if len(parts) < 2:
        continue

    try:
        lat = float(parts[0])
        lon = float(parts[1])
    except ValueError:
        continue

    feature = {
        "type": "Feature",
        "geometry": {
            "type": "Point",
            "coordinates": [lon, lat]
        },
        "properties": {
            "submission_id": record.get("_id"),
            "flood_depth_m": record.get("Flood_depth_m", ""),
            "date_observed": record.get("Date_observed", ""),
            "ward_name": record.get("Ward_name", ""),
            "households": record.get("Affected_households", ""),
            "water_source": record.get("Water_source", ""),
            "damage": record.get("Infrastructure_damage", ""),
            "notes": record.get("Observer_notes", ""),
        }
    }
    features.append(feature)

geojson = {
```

```
    "type": "FeatureCollection",
    "name": "tanga_flood_survey",
    "crs": {"type": "name", "properties": {"name": "urn:ogc:def:crs:OGC:1.3:CRS84"}},
    "features": features
}

# Save to file
with open("tanga_flood_survey.geojson", "w") as f:
    json.dump(geojson, f, indent=2)

print(f"Saved {len(features)} features to tanga_flood_survey.geojson")
```

Step 4: Upload to Local CRD

1. Make sure your local CRD is running: `cd  /my-crd && docker compose up -d`
2. Open `http://localhost` and log in as admin
3. Click **Data** → **Upload Layer**
4. Select `tanga_flood_survey.geojson`
5. Set metadata:
    - Title: `Tanga Flood Survey 2024`
    - Keywords: `tanga, flood, survey, kobo, gis`
    - Category: Hazard
6. Click **Upload**

Step 5: View Tanga Data in CRD Map

```
# In CRD Maps:
# 1. Create a new map
# 2. Add Tanga_imagery_reduced (from live CRD via WMS)
# WMS URL: https://crd.resilienceacademy.ac.tz/geoserver/ows
# 3. Add your tanga_flood_survey layer (from local CRD)
# 4. Click a flood survey point to see attribute popup
# 5. Save the map
```

Checkpoint:

✓ KoboToolbox API returned your Tanga flood submissions
✓ `tanga_flood_survey.geojson` saved with correct feature count
✓ Layer uploaded successfully to local CRD
✓ Tanga flood points visible on CRD map
✓ Tanga satellite imagery (`Tanga_imagery_reduced`) visible as WMS background